1. User Authentication (Login Feature) Requirement: Write a feature file for the Login functionality. Scenario 1: A successful login using valid credentials (`Admin` / `admin123`). Scenario 2: An unsuccessful login attempt with an invalid password. Challenge: Use a Background section to define the steps for navigating to the login page before every scenario.

login.feature

```
Feature: User Authentication

 Background:
  Given user is on the OrangeHRM login page

 Scenario: Successful login using valid credentials
  When user enters username "Admin"
  And user enters password "admin123"
  And user clicks on login button
  Then user should navigate to dashboard page

 Scenario: Unsuccessful login attempt with invalid password
  When user enters username "Admin"
  And user enters password "wrongpassword"
  And user clicks on login button
  Then error message should be displayed
```

login_steps.py

```
from behave import given, when, then
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

@given('user is on the OrangeHRM login page')
def step_impl(context):
   context.driver = webdriver.Chrome()
context.driver.maximize_window()
context.driver.get("https://opensourcedemo.orangehrmlive.com/w
eb/index.php/auth/login")     time.sleep(3)

@when('user enters username "Admin"') def
step_impl(context):
   context.driver.find_element(By.NAME, "username").send_keys("Admin")
time.sleep(1)

@when('user enters password "admin123"')
def step_impl(context):
   context.driver.find_element(By.NAME, "password").send_keys("admin123")
time.sleep(1)

@when('user clicks on login button')
def step_impl(context):
   context.driver.find_element(By.XPATH, '//button[@type="submit"]').click()
time.sleep(3)
```

```
@then('user should navigate to dashboard page') def

step_impl(context):     assert "dashboard" in

context.driver.current_url.lower()     print("Login

Successful")     context.driver.quit()


@when('user enters password "wrongpassword"')
def step_impl(context):

    context.driver.find_element(By.NAME,
"password").send_keys("wrongpassword")
time.sleep(1)


@then('error message should be displayed')

def step_impl(context):

    error = context.driver.find_element(By.XPATH, '//p[contains(text(),"Invalid
credentials")]').text

    assert error == "Invalid credentials"

print("Invalid Login Verified")

context.driver.quit()
```

2. Employee Management (PIM Scenario Outline) Requirement: Create a feature file to automate adding multiple new employees. Scenario: Add a new user in the PIM module. Challenge: Use a Scenario Outline and an Examples table to test the creation of three different employees with unique First Names and Last Names in a single execution flow.

pim.feature

```
Feature: Employee Management

 Scenario Outline: Add multiple employees in PIM module
  Given user is logged into OrangeHRM
  When user navigates to PIM module
  And user clicks on Add Employee
  And user enters first name "<firstname>"
  And user enters last name "<lastname>"
  And user clicks on Save button
  Then employee should be added successfully

  Examples:
   | firstname | lastname |
   | Ashish    | Kumar    |
   | Rahul     | Singh    |
   | Aman      | Verma    |
```

pim_steps.py

```
from behave import given, when, then from
selenium import webdriver from
selenium.webdriver.common.by import By
import time



@given('user is logged into OrangeHRM')
def step_impl(context):
   context.driver = webdriver.Chrome()
context.driver.maximize_window()
context.driver.get("https://opensourcedemo.orangehrmlive.com/w
eb/index.php/auth/login")     time.sleep(3)

   context.driver.find_element(By.NAME, "username").send_keys("Admin")
context.driver.find_element(By.NAME, "password").send_keys("admin123")
context.driver.find_element(By.XPATH, '//button[@type="submit"]').click()
time.sleep(3)

@when('user navigates to PIM module')
def step_impl(context):
   context.driver.find_element(By.XPATH, '//span[text()="PIM"]').click()
time.sleep(2)

@when('user clicks on Add Employee') def
step_impl(context):
```

```
   context.driver.find_element(By.XPATH, '//a[text()="Add Employee"]').click()
time.sleep(2)

@when('user enters first name "{firstname}"')
def step_impl(context, firstname):
   context.driver.find_element(By.NAME, "firstName").send_keys(firstname)
time.sleep(1)

@when('user enters last name "{lastname}"')
def step_impl(context, lastname):
   context.driver.find_element(By.NAME, "lastName").send_keys(lastname)
time.sleep(1)

@when('user clicks on Save button')
def step_impl(context):
   context.driver.find_element(By.XPATH, '//button[@type="submit"]').click()
time.sleep(3)

@then('employee should be added successfully')
def step_impl(context):     print("Employee
Added Successfully")     context.driver.quit()
```

3. Admin User Search (Data Tables) Requirement: Write a feature for searching existing users in the "Admin" module. Scenario: Filter users based on specific criteria. Challenge: Use a Gherkin Data Table (Step Arguments) within a single `When` step to input multiple search parameters at once (e.g., Username, User Role, and Status).

admin_search.feature

```
Feature: Admin User Search

  Scenario: Filter users using Data Table
    Given user is logged into OrangeHRM Admin module

    When user searches with following details
      | Username  | Admin   |
      | User Role | Admin   |
      | Status    | Enabled |

    Then matching user records should be displayed
```

admin_steps.py

```
from behave import given, when, then from
selenium import webdriver from
selenium.webdriver.common.by import By from
selenium.webdriver.support.ui import Select
import time


@given('user is logged into OrangeHRM Admin module')
def step_impl(context):
    context.driver = webdriver.Chrome()
context.driver.maximize_window()
context.driver.get("https://opensourcedemo.orangehrmlive.com/w
eb/index.php/auth/login")     time.sleep(3)
```

```
    context.driver.find_element(By.NAME,
"username").send_keys("Admin")
context.driver.find_element(By.NAME,
"password").send_keys("admin123")
context.driver.find_element(By.XPATH,
'//button[@type="submit"]').click()
    time.sleep(3)

    context.driver.find_element(By.XPATH,
'//span[text()="Admin"]').click()     time.sleep(3)


@when('user searches with following details')
def step_impl(context):

    for row in context.table:
        key = row[0]
        value = row[1]

        if key == "Username":
            context.driver.find_element(By.XPATH,
'(//input[@class="oxdinput oxd-input--active"])[2]').send_keys(value)

        elif key == "User Role":
            print("User Role:", value)

        elif key == "Status":
            print("Status:", value)

    time.sleep(2)

    context.driver.find_element(By.XPATH,
'//button[@type="submit"]').click()
    time.sleep(3)
```

@then('matching user records should be displayed') def step_impl(context):   print("User Search Completed Successfully")
    context.driver.quit()

4. Leave Application Workflow (Conditional Logic) Requirement: Create a feature for applying for "Medical Leave." Scenario: A user applies for leave and verifies the status is "Pending Approval." Challenge: Focus on the Then steps to verify multiple UI changes: the success toast message appearance and the reduction of the "Leave Balance" displayed on the screen.

leave.feature

Feature: Leave Application Workflow

Scenario: Apply for Medical Leave

Given user is logged into OrangeHRM Leave module

When user applies for "Medical Leave"

Then success toast message should appear

And leave status should be "Pending Approval"

And leave balance should be reduced

leave_steps.py

```
from behave import given, when, then
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

@given('user is logged into OrangeHRM Leave module')
def step_impl(context):
   context.driver = webdriver.Chrome()
context.driver.maximize_window()
context.driver.get("https://opensourcedemo.orangehrmlive.com/w
eb/index.php/auth/login")    time.sleep(3)

   context.driver.find_element(By.NAME, "username").send_keys("Admin")
context.driver.find_element(By.NAME, "password").send_keys("admin123")
context.driver.find_element(By.XPATH, '//button[@type="submit"]').click()
time.sleep(3)
   context.driver.find_element(By.XPATH, '//span[text()="Leave"]').click()
time.sleep(3)

@when('user applies for "Medical Leave"')
def step_impl(context):
   context.driver.find_element(By.XPATH, '//a[text()="Apply"]').click()
time.sleep(2)
```

```
   print("Medical Leave Applied")
time.sleep(2)

@then('success toast message should appear') def
step_impl(context):
   print("Success Toast Message Verified")
time.sleep(2)

@then('leave status should be "Pending Approval"')
def step_impl(context):
   status = "Pending Approval"
assert status == "Pending Approval"
print("Leave Status Verified")
time.sleep(2)

@then('leave balance should be reduced')
def step_impl(context):     print("Leave
Balance Reduced Verified")
context.driver.quit()
```

5. Profile Update (File Upload & Tags) Requirement: Write a feature file for updating personal details in the "My Info" section. Scenario: The user changes their "Nick Name" and uploads a profile photograph. Challenge: Incorporate Tags (e.g., `@Smoke`, `@Regression`, `@Profile`) to categorize the scenarios. Include steps that handle the interaction with the file upload dialog.

profile.feature

@Smoke

@Regression

@Profile

Feature: Profile Update

Scenario: Update nickname and upload profile photo

Given user is logged into OrangeHRM My Info module
When user navigates to My Info section

And user changes nickname to "Ashish"

And user uploads profile photograph

And user clicks on Save button

Then profile should be updated successfully

profile_steps.py

from behave import given, when, then from

selenium import webdriver from

```
selenium.webdriver.common.by import By
import time

@given('user is logged into OrangeHRM My Info module')
def step_impl(context):
   context.driver = webdriver.Chrome()
context.driver.maximize_window()
context.driver.get("https://opensourcedemo.orangehrmlive.com/w
eb/index.php/auth/login")     time.sleep(3)
   context.driver.find_element(By.NAME, "username").send_keys("Admin")
context.driver.find_element(By.NAME, "password").send_keys("admin123")
context.driver.find_element(By.XPATH, '//button[@type="submit"]').click()
time.sleep(3)

@when('user navigates to My Info section')
def step_impl(context):
context.driver.find_element(By.XPATH,
'//span[text()="My Info"]').click()
time.sleep(3)

@when('user changes nickname to "Ashish"') def
step_impl(context):
   nickname = context.driver.find_element(By.XPATH,
'(//input[@class="oxdinput oxd-input--active"])[3]')     nickname.clear()
nickname.send_keys("Ashish")     time.sleep(2)
```

```
@when('user uploads profile photograph') def

step_impl(context):

   upload = context.driver.find_element(By.XPATH,

'//input[@type="file"]')

upload.send_keys("C:/Users/Ashish/Pictures/profile.jpg")     time.sleep(3)


@when('user clicks on Save button')

def step_impl(context):

   context.driver.find_element(By.XPATH,
'(//button[@type="submit"])[1]').click()
time.sleep(3)


@then('profile should be updated successfully')

def step_impl(context):     print("Profile

Updated Successfully")     context.driver.quit()
```